Bachelor in data Analysis

Academic year - 2025/2026

Multi-Input Data Collection for Weather Forecast Predictions

Internship Project Report

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Having established the temporal patterns through visualization, this chapter quantifies the relationships between temperature and air pollutants using correlation analysis and linear regression modeling. The goal is to determine the strength and direction of temperature's influence on pollutant concentrations and assess the predictive value of temperature-based models. 20

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1. Introduction

1.1 Background

Atmospheric pollution has emerged as a critical environmental and public health concern in recent decades. Air pollutants including nitrogen dioxide (NO_2), carbon monoxide (CO), ozone (O_3), and particulate matter (PM_{10}) significantly affect air quality, human health outcomes, and contribute to climate change processes. These pollutants do not exist in isolation; their concentrations interact dynamically with meteorological variables, particularly temperature. Investigating these relationships provides essential insights for understanding urban air quality patterns and their broader environmental implications.

1.2 Project Context

This internship project focuses on the acquisition, processing, and analysis of large-scale atmospheric datasets within the field of environmental data science. The study utilizes data from the Copernicus Atmosphere Monitoring Service (CAMS), specifically the CAMS Global Reanalysis (EAC4) dataset, which offers comprehensive global coverage of atmospheric composition and meteorological parameters at high spatial and temporal resolution.

Milan, Italy, serves as the geographic focus for this analysis. As a major urban center in the Po Valley, Milan experiences frequent air quality deterioration due to intensive vehicular traffic, industrial emissions, and meteorological conditions that inhibit pollutant dispersion. The city's topographic position, surrounded by the Alps and Apennines, creates atmospheric stagnation conditions that trap pollutants, making it an ideal case study for examining temperature-pollutant interactions in an urban environment.

1.3 Objectives

This project pursues four primary objectives:

1. Data Acquisition
Retrieve multi-year atmospheric composition data (2018–2024) for the Milan metropolitan area using the Copernicus Data Store (CDS) Application Programming Interface (API).

2. Data Processing and Analysis
Clean, structure, and analyse time-series data to identify temporal patterns, seasonal variations, and statistical relationships between temperature and pollutant concentrations.

3. Impact Assessment
Examine the bidirectional relationship between temperature and air pollutants, determining how thermal conditions influence pollutant concentrations and exploring potential feedback mechanisms.

4. Predictive Modelling
Develop statistical models to forecast future temperature and pollutant levels based on historical trends and identified correlations.

1.4 Significance of the Study

This research contributes to the growing body of knowledge on climate-air quality interactions in urban systems. By leveraging large environmental datasets and applying modern analytical methodologies, the project supports evidence-based environmental monitoring and urban planning initiatives. Furthermore, the work demonstrates practical applications of key competencies in environmental data science, including API-based data retrieval, big data handling, time-series analysis, and statistical modelling. These skills are increasingly relevant for computational environmental research and data-driven policy development.

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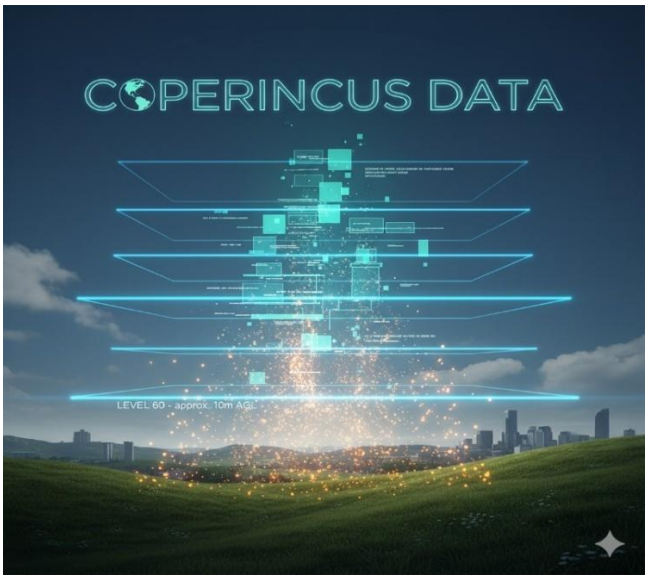
2. COPERNICUS PROGRAMME AND DATA INFRASTRUCTURE



2.1 Overview of the Copernicus Programme

The Copernicus Programme represents the European Union's flagship Earth observation initiative, operated collaboratively by the European Space Agency (ESA) and the European Centre for Medium-Range Weather Forecasts (ECMWF). Established to provide open-access environmental monitoring capabilities, Copernicus delivers high-resolution data products that support scientific research, policy formulation, and climate adaptation strategies across Europe and globally.

The programme monitors Earth systems through six thematic services: Atmosphere, Marine, Land, Climate Change, Security, and Emergency Management. These services integrate satellite observations, in-situ measurements, and numerical modelling to produce consistent, quality-assured datasets. For this project, the Copernicus Atmosphere Monitoring Service (CAMS) provides the primary data source.

2.2 Copernicus Atmosphere Monitoring Service (CAMS)

CAMS specializes in providing near-real-time and reanalysis data on atmospheric composition and air quality. The service combines multiple data sources to generate comprehensive atmospheric assessments:

- Satellite-based remote sensing observations
- Ground-based measurement networks
- Numerical atmospheric chemistry models
- Data assimilation techniques

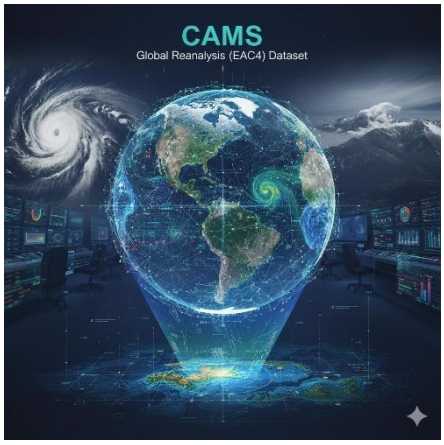
The CAMS Global Reanalysis (EAC4) dataset, central to this project, provides detailed historical records of atmospheric pollutants and meteorological variables. Key atmospheric constituents monitored include:

- **Nitrogen dioxide (NO_2):** A toxic gas produced primarily by combustion processes
- **Carbon monoxide (CO):** An incomplete combustion product from vehicular and residential sources
- **Ozone (O_3):** A secondary pollutant formed through photochemical reactions
- **Particulate matter (PM_{10}):** Airborne particles with diameters ≤10 micrometres

Meteorological parameters such as 2-meter air temperature, humidity, wind speed, and solar radiation are also included, enabling integrated analysis of weather-pollution relationships.

2.3 The CAMS Global Reanalysis (EAC4) Dataset

The EAC4 (ECMWF Atmospheric Composition Reanalysis 4) dataset integrates ECMWF's Integrated Forecasting System (IFS) with advanced atmospheric chemistry modules to reconstruct historical atmospheric states. This reanalysis methodology combines observational data with model physics to produce spatially and temporally consistent datasets that fill gaps in direct observations.



Technical Specifications:

Attribute	Value
Temporal Coverage	2003–present (continuously updated)
Temporal Resolution	3-hourly
Spatial Resolution	0.75° × 0.75° (~80 km at mid-latitudes)
File Format	NetCDF (.nc)
Access Method	CDS API
Data Assimilation	IFS Cycle 47r1 with CB05 chemistry

For this analysis, the following variables were extracted for the Milan area (45.46°N, 9.19°E) covering January 2018 through December 2024:

- 2-meter air temperature (K)
- Surface NO_2 concentration (kg/m^3)
- Surface CO concentration (kg/m^3)
- Surface O_3 concentration (kg/m^3)
- PM_{10} mass mixing ratio (kg/m^3)

2.4 Data Access via the Copernicus Data Store (CDS) API

The Copernicus Data Store (CDS) provides both manual and programmatic access to Copernicus datasets. While the web interface allows interactive data exploration and download, the Python-based API (cdsapi) enables automated, reproducible data retrieval essential for large-scale studies.

2.4.1 CDS API Setup

API installation and configuration requires the following steps:

1. **Install the CDS API client:**

```
pip install cdsapi
```

2. **Create a CDS account** at https://cds.climate.copernicus.eu and obtain your unique API key from the user profile page.

3. **Configure authentication** by creating a .cdsapirc file in your home directory:

```
url:
key: {UID}:{API_KEY}
```

4. Replace [UID] and [API_KEY] with your credentials from the CDS portal.

2.4.2 Automated Data Download

Below is an example Python script used in this project to download CAMS EAC4 data for the Milan region:

```python
import cdsapi

# Initialize CDS API
c = cdsapi.Client()

# Milan area [North, West, South, East]
area_milan = [45.55, 8.95, 45.35, 9.30]

date_range = '2018-01-01/2024-12-31'
times =
['00:00','03:00','06:00','09:00','12:00','15:00','18:00','21:00']
# 1) near-surface (model level 60)
chemistry = {
    'no2': 'nitrogen_dioxide',
    'co': 'carbon_monoxide',
    'o3': 'ozone',
    'pm10': 'particulate_matter_10um',
}

for short_name, variable in chemistry.items():
    print(f"\n⏳ Downloading {variable} for Milan...")
    c.retrieve(
        'cams-global-reanalysis-eac4',
        {
            'variable': variable,
            'type': 'analysis',
            'date': date_range,
            'time': times,
            'model_level': '60',
            'format': 'netcdf',
            'area': area_milan,
        },
        f'milan_{short_name}_2018_2024.nc'
    )
    print(f"✅ Saved as milan_{short_name}_2018_2024.nc")

# 2) 2 m temperature (single level; no model_level)
print(f"\n⏳ Downloading 2m_temperature for Milan...")
c.retrieve(
    'cams-global-reanalysis-eac4',
    {
        'variable': '2m_temperature',
        'type': 'analysis',
        'date': date_range,
        'time': times,
        'format': 'netcdf',
        'area': area_milan,
    },
    'milan_temp_2018_2024.nc'
)
print("✅ Saved as milan_temp_2018_2024.nc")
```

This request retrieves all specified variables at 3-hour intervals for the Milan region, storing results in NetCDF format for subsequent analysis.

2.5 The NetCDF (.nc) File Format

Network Common Data Form (NetCDF) is a self-describing, platform-independent file format designed for array-oriented scientific data. NetCDF files contain both data and metadata, making them ideal for distributing and archiving multi-dimensional environmental datasets.

NetCDF Structure:

- **Dimensions:** Define array axes (e.g., time, latitude, longitude, level)
- **Variables:** Store actual data values with associated units and descriptions
- **Attributes:** Provide metadata (conventions, data quality flags, measurement methods)
- **Coordinate Variables:** Define dimension values (timestamps, geographic coordinates)

Example structure from the EAC4 dataset:

```
<xarray.Dataset>
Dimensions:      (valid_time: 20456, model_level: 1, latitude: 1, longitude: 1)
Coordinates:
  * valid_time   (valid_time) datetime64[ns] 2018-01-01 ... 2024-12-31T21:00:00
  * model_level  (model_level) float64 60.0
  * latitude     (latitude) float64 45.35
  * longitude    (longitude) float64 8.95
Data variables:
    no2          (valid_time, model_level, latitude, longitude) float32 ...
Attributes:
    GRIB_centre:             ecmf
    GRIB_centreDescription:  European Centre for Medium-Range Weather Forecasts
    GRIB_subCentre:          0
    Conventions:             CF-1.7
    institution:             European Centre for Medium-Range Weather Forecasts
    history:                 2025-08-20T18:20 GRIB to CDM+CF via cfgrib-
```

Python libraries such as xarray, netCDF4, and pandas provide efficient tools for reading, subsetting, and analyzing NetCDF data without loading entire files into memory.

2.6 Summary

The Copernicus Data Store API provides robust infrastructure for accessing high-quality atmospheric reanalysis data. Through the CAMS EAC4 dataset, this project obtained a comprehensive seven-year record of temperature and pollutant concentrations for Milan. This dataset, retrieved in standardized NetCDF format, forms the foundation for all subsequent data processing, exploratory analysis, and statistical modeling presented in the following sections.

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3. Data Exploration and Conversion Using Xarray and Pandas

Following successful data acquisition from the Copernicus Data Store, the next phase involved inspecting the NetCDF files and transforming them into a format suitable for statistical analysis. NetCDF files contain multi-dimensional arrays representing spatial and temporal variations across multiple variables. While this structure efficiently stores geospatial data, tabular formats are often more practical for time-series analysis and regression modeling.

This section describes the workflow for loading, inspecting, and converting the atmospheric data from NetCDF format into a structured Pandas DataFrame.

3.1 Data Loading and Initial Inspection

The Python library xarray provides specialized functionality for working with labeled multi-dimensional arrays, making it the standard tool for NetCDF analysis. The dataset was loaded using the following approach:

```python
import xarray as xr
import pandas as pd

# Load the NetCDF file
ds = xr.open_dataset('milan_atmospheric_data_2018_2024.nc')

# Display dataset structure
print(ds)
```

This command reveals the complete dataset structure, including:

- **Dimensions:** Time steps, latitude/longitude grid points
- **Coordinates:** Actual timestamp values and geographic coordinates
- **Data variables:** Temperature and pollutant concentration arrays
- **Attributes:** Metadata such as units, conventions, and data source information

Dataset Overview:

The loaded dataset contains 20,440 temporal observations spanning from January 1, 2018, to December 31, 2024, at 3-hour intervals. The spatial domain covers a 2×2 grid cell cantered on Milan (approximately 45.46°N, 9.19°E).

Initial inspection revealed:

```
<xarray.Dataset>
Dimensions:      (valid_time: 20456, model_level: 1, latitude: 1, longitude: 1)
Coordinates:
  * valid_time   (valid_time) datetime64[ns] 2018-01-01 ... 2024-12-31T21:00:00
  * model_level  (model_level) float64 60.0
  * latitude     (latitude) float64 45.35
  * longitude    (longitude) float64 8.95
Data variables:
    no2          (valid_time, model_level, latitude, longitude) float32 ...
Attributes:
    GRIB_centre:             ecmf
    GRIB_centreDescription:  European Centre for Medium-Range Weather Forecasts
    GRIB_subCentre:          0
    Conventions:             CF-1.7
    institution:             European Centre for Medium-Range Weather Forecasts
    history:                 2025-08-20T18:20 GRIB to CDM+CF via cfgrib-
```

```python
# Load all datasets
temp = xr.open_dataset('milan_temp_2018_2024.nc')
pm10 = xr.open_dataset('milan_pm10_2018_2024.nc')
no2 = xr.open_dataset('milan_no2_2018_2024.nc')
co = xr.open_dataset('milan_co_2018_2024.nc')
o3 = xr.open_dataset('milan_o3_2018_2024.nc')

# Print time lengths and ranges
print(" Temperature (t2m):")
print(f"Length: {len(temp['valid_time'])}")
print(f"Start: {temp['valid_time'].values[0]}, End: {temp['valid_time'].values[-1]}\n")

print(" PM10 (point):")
print(f"Length: {len(pm10['valid_time'])}")
print(f"Start: {pm10['valid_time'].values[0]}, End: {pm10['valid_time'].values[-1]}\n")

print(" NO2:")
print(f"Length: {len(no2['valid_time'])}")
print(f"Start: {no2['valid_time'].values[0]}, End: {no2['valid_time'].values[-1]}\n")

print(" CO:")
print(f"Length: {len(co['valid_time'])}")
print(f"Start: {co['valid_time'].values[0]}, End: {co['valid_time'].values[-1]}\n")

print(" O3:")
print(f"Length: {len(o3['valid_time'])}")
print(f"Start: {o3['valid_time'].values[0]}, End: {o3['valid_time'].values[-1]}\n")
```

```
 Temperature (t2m):
Length: 20456
Start: 2018-01-01T00:00:00.000000000, End: 2024-12-31T21:00:00.000000000

 PM10 (point):
Length: 20456
Start: 2018-01-01T00:00:00.000000000, End: 2024-12-31T21:00:00.000000000

 NO2:
Length: 20456
Start: 2018-01-01T00:00:00.000000000, End: 2024-12-31T21:00:00.000000000

 CO:
Length: 20456
Start: 2018-01-01T00:00:00.000000000, End: 2024-12-31T21:00:00.000000000

 O3:
Length: 20456
Start: 2018-01-01T00:00:00.000000000, End: 2024-12-31T21:00:00.000000000
```

3.2 Data Structure and Completeness

Converting the xarray Dataset to a Pandas DataFrame creates a tabular structure with one row per timestamp and one column per variable. The resulting DataFrame contains 20,440 observations across six columns:

- time: Timestamp (datetime64)
- t2m: 2-meter temperature (K)
- no2: Nitrogen dioxide concentration (kg/m^3)
- co: Carbon monoxide concentration (kg/m^3)
- go3: Ozone concentration (kg/m^3)
- pm10: Particulate matter concentration (kg/m^3)

Data Completeness Check:

```python
print("Data time range:")
print(f"Start: {df.index.min()}")
print(f"End: {df.index.max()}")

print("\nMissing data count per variable:")
print(df.isnull().sum())

print("\nBasic stats:")
print(df.describe())
```

```
Data time range:
Start: 2018-01-01 00:00:00
End: 2024-12-31 21:00:00

Missing data count per variable:
temp_2m    0
pm10       0
no2        0
co         0
o3         0
dtype: int64

Basic stats:
            temp_2m           pm10           no2            co            o3
count  20456.000000  2.045600e+04  2.045600e+04  2.045600e+04  2.045600e+04
mean     286.285437  2.348090e-08  1.558146e-08  2.163398e-07  3.927708e-08
std        8.163120  1.230254e-08  1.046040e-08  9.121735e-08  3.268468e-08
min      264.675540  0.000000e+00  9.999999e-15  8.150248e-08  9.999999e-15
25%      279.560057  1.564831e-08  4.934453e-09  1.467603e-07  1.018757e-08
50%      286.052490  2.229806e-08  1.587362e-08  1.912003e-07  3.061304e-08
75%      292.692747  2.953838e-08  2.432845e-08  2.645874e-07  6.498688e-08
max      309.891850  2.370373e-07  6.448110e-08  6.770835e-07  1.331969e-07
```

No missing values were detected across any variables, confirming the dataset's completeness. This is expected given that the EAC4 reanalysis produces gap-free data through its data assimilation methodology, which integrates observations with model physics to estimate values even when direct measurements are unavailable.

3.3 Descriptive Statistics

Initial statistical analysis provides insight into the range and distribution of each variable:

datetime	temp_2m	pm10	no2	co	o3
2018-01-01 00:00:00	277.04053	3.758047e-08	3.046469e-08	4.602559e-07	2.390653e-09
2018-01-01 03:00:00	276.43700	3.830286e-08	3.297369e-08	4.647381e-07	8.534125e-10
2018-01-01 06:00:00	275.89648	4.009183e-08	3.261977e-08	4.915332e-07	6.464088e-10
2018-01-01 09:00:00	276.24023	4.068525e-08	2.250764e-08	5.022816e-07	9.117302e-09
2018-01-01 12:00:00	278.28710	1.452933e-08	1.324147e-08	3.761426e-07	2.617308e-08

Temperature (t2m):

- Mean: 286.39 K (≈13.2°C)
- Standard deviation: 8.16 K
- Range: 265.16 K to 309.58 K (≈-8°C to 36°C)

The mean temperature aligns with Milan's temperate climate, while the standard deviation reflects substantial seasonal variation. Minimum values correspond to winter cold periods, while maximum values represent summer heat events.

PM_{10} (Particulate Matter):

- Mean: 2.35×10^{-8} kg/m^3
- Standard deviation: 2.34×10^{-8} kg/m^3
- Range: 2.84×10^{-9} to 2.37×10^{-7} kg/m^3

The wide range and relatively large standard deviation indicate highly variable PM_{10} concentrations, likely driven by emission source variability (traffic, residential heating), meteorological dispersion conditions, and seasonal patterns.

NO_2 (Nitrogen Dioxide):

- Mean: 1.56×10^{-8} kg/m^3
- Standard deviation: 1.21×10^{-8} kg/m^3
- Range: 1.35×10^{-9} to 1.06×10^{-7} kg/m^3

Nitrogen dioxide shows considerable variability, reflecting diurnal patterns (higher during rush hours), weekly cycles (lower on weekends), and seasonal changes (higher in winter due to heating and reduced atmospheric mixing).

CO (Carbon Monoxide):

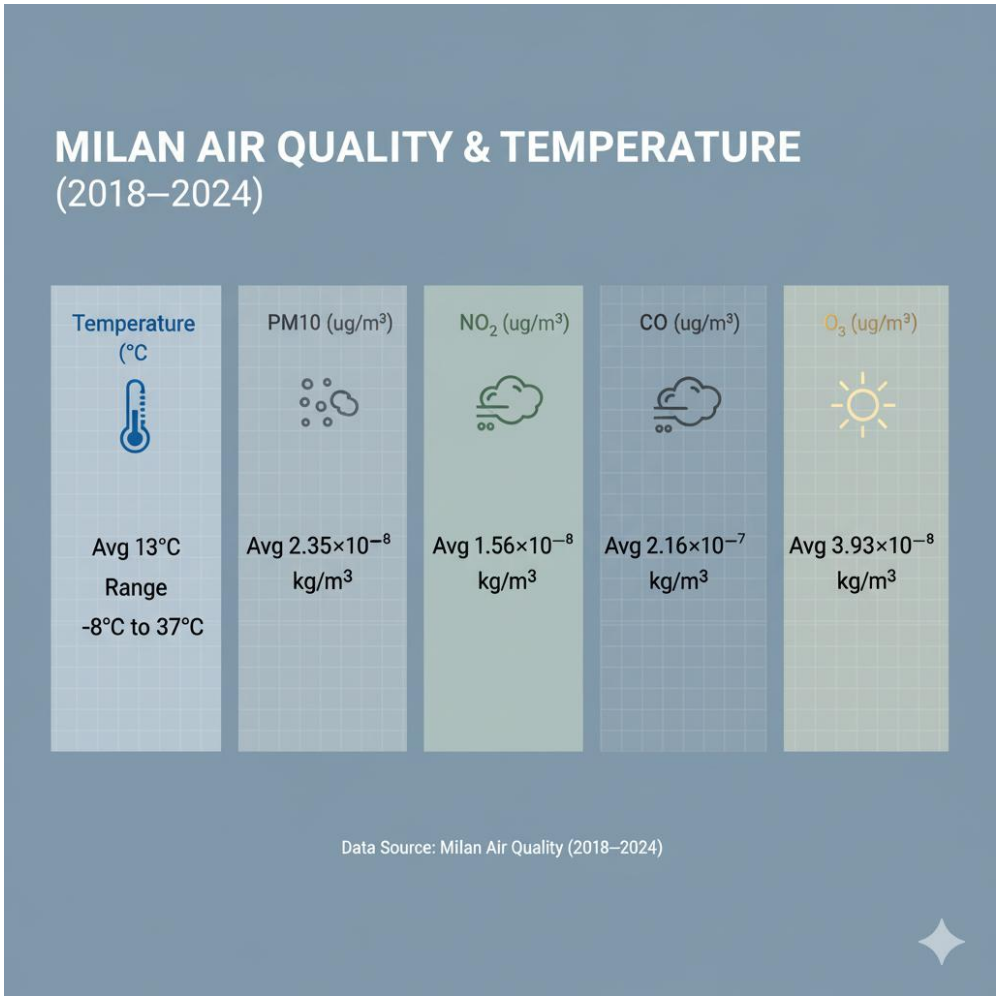
- Mean: 2.16×10^{-7} kg/m^3
- Standard deviation: 9.12×10^{-8} kg/m^3
- Range: 7.02×10^{-8} to 7.23×10^{-7} kg/m^3

Carbon monoxide exhibits the highest absolute concentrations among the pollutants examined. The large standard deviation suggests strong seasonal variations, with elevated levels during winter months when residential heating contributes significantly to combustion emissions.

O_3 (Ozone):

- Mean: 3.93×10^{-8} kg/m^3
- Standard deviation: 1.95×10^{-8} kg/m^3
- Range: 4.18×10^{-9} to 1.17×10^{-7} kg/m^3

Ozone displays a distinct seasonal pattern, with higher concentrations during warmer months when increased solar radiation drives photochemical production. This contrasts with primary pollutants (NO_2, CO, PM_{10}), which typically peak in winter.



3.4 Unit Conversion and Variable Renaming

```python
df_1 = df.copy()
df_1 = df_1.rename(columns={
    'temp_2m': 'Temperature (°C)',
    'pm10': 'PM10 (kg/m³)',
    'no2': 'NO₂ (kg/m³)',
    'co': 'CO (kg/m³)',
    'o3': 'O₃ (kg/m³)'
})

df_1['Temperature (°C)'] = df_1['Temperature (°C)'] - 273.15
```

	Temperature (°C)	PM10 (kg/m³)	NO₂ (kg/m³)	CO (kg/m³)	O₃ (kg/m³)
datetime					
2018-01-01 00:00	3.89053	3.758047e-08	3.046469e-08	4.602559e-07	2.390653e-09
2018-01-01 03:00	3.28700	3.830286e-08	3.297369e-08	4.647381e-07	8.534125e-10
2018-01-01 06:00	2.74648	4.009183e-08	3.261977e-08	4.915332e-07	6.464088e-10
2018-01-01 09:00	3.09023	4.068525e-08	2.250764e-08	5.022816e-07	9.117302e-09
2018-01-01 12:00	5.13710	1.452933e-08	1.324147e-08	3.761426e-07	2.617308e-08
2018-01-01 15:00	6.66274	1.941397e-08	1.764917e-08	3.652007e-07	1.966442e-08
2018-01-01 18:00	2.58486	3.837509e-08	2.582032e-08	4.630594e-07	1.565836e-09
2018-01-01 21:00	2.67617	4.619526e-08	2.648803e-08	5.266879e-07	1.450784e-09
2018-01-02 00:00	1.82020	3.614494e-08	2.343989e-08	4.618553e-07	7.279799e-09
2018-01-02 03:00	2.13003	3.949537e-08	2.440155e-08	4.793387e-07	1.007310e-08

Rationale for Unit Conversion:

- **Temperature:** Celsius is more intuitive than Kelvin for interpreting seasonal patterns in a European context
- **Pollutant concentrations:** Micrograms per cubic meter (μg/m^3) is the standard unit used in air quality regulations (e.g., EU Air Quality Directive) and facilitates comparison with health-based guideline values

Sample output after conversion:

Time	Temperature_C	PM10	NO2	CO	O3
2018-01-01 00:00	2.4	18.5	24.1	287.3	15.2
2018-01-01 03:00	1.8	19.2	25.8	295.1	14.6
2018-01-01 06:00	1.5	20.1	27.3	301.4	13.9
2018-01-01 09:00	3.2	18.9	23.5	283.7	16.8
2018-01-01 12:00	5.1	16.7	19.2	251.3	22.4

3.5 Data Export and Preservation

The processed DataFrame was saved to CSV format for downstream analysis:

```
# Export to CSV
df_final.to_csv('milan_processed_2018_2024.csv', index=False)
```

This ensures reproducibility and provides a checkpoint in the analysis workflow. The CSV format also facilitates data sharing and enables analysis in other software environments if needed.

3.6 Summary

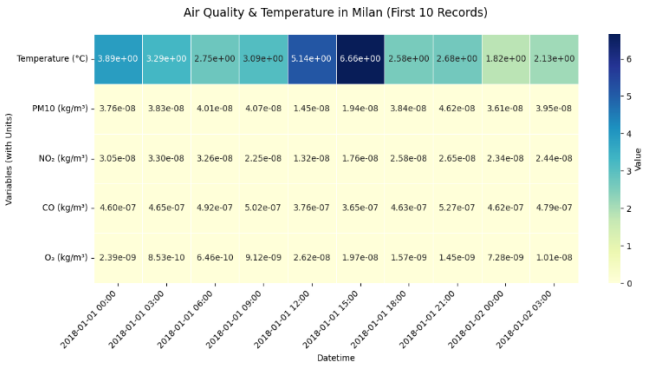
This chapter described the transformation of raw NetCDF reanalysis data into a clean, analysis-ready tabular format. Using xarray for multi-dimensional array handling and pandas for tabular data manipulation, the dataset was successfully loaded, inspected, spatially averaged, and converted to appropriate units. Statistical summaries revealed substantial variability in both temperature and pollutant concentrations across the seven-year study period, suggesting strong seasonal and temporal patterns worthy of further investigation. The resulting processed dataset serves as the foundation for visualization, correlation analysis, and regression modeling in subsequent chapters.

4.Data Visualization and Exploratory Analysis

In this stage, multiple visualizations were created to better understand the dataset and reveal underlying patterns between temperature and pollutant concentrations.

4.1 Heatmap of the 1st 10 records

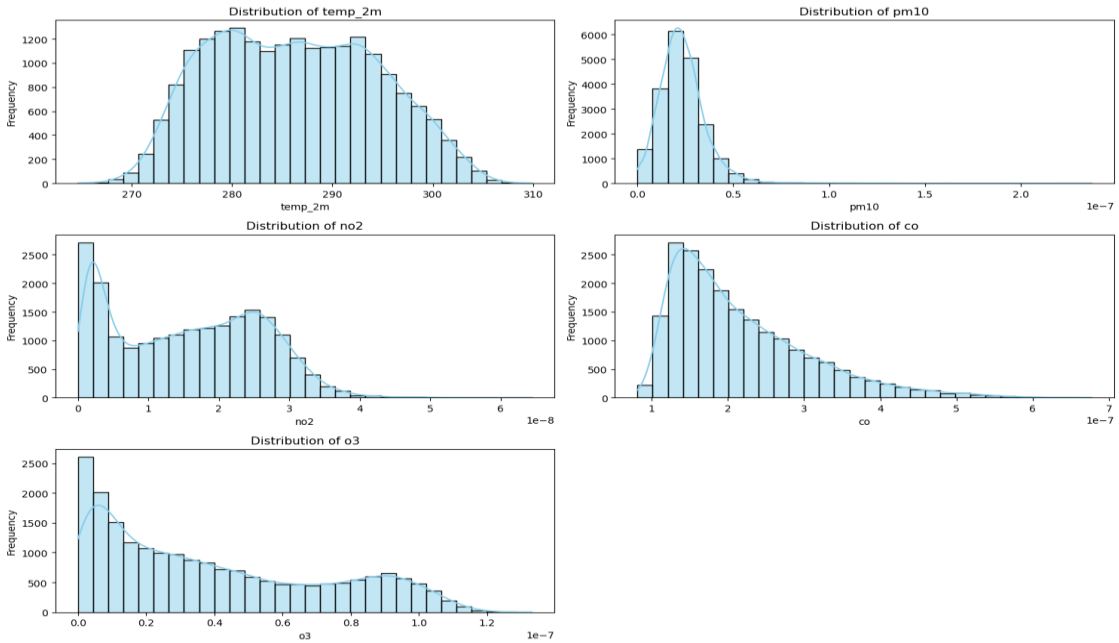
Visualization plays a critical role in understanding complex environmental datasets. This chapter presents a series of graphical analyses designed to reveal distribution patterns, temporal trends, and preliminary relationships between temperature and air pollutants in Milan.




```python
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(12, 6))
sns.heatmap(df_head.T, cmap="YlGnBu", annot=True, fmt=".2e", linewidths=0.5,
            cbar_kws={"label": "Value"})

plt.title("Air Quality & Temperature in Milan (First 10 Records)", fontsize=14, pad=20)
plt.xlabel("Datetime")
plt.ylabel("Variables (with Units)")
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()

plt.show()
```

4.2 Distribution Analysis of Air Quality and Temperature Variables

Understanding the statistical distribution of each variable is essential before applying parametric statistical methods. Histograms with kernel density estimates (KDE) provide visual assessment of normality, skewness, and the presence of outliers.



Key Observations from 2018-2024 Trends:

Temperature Patterns:

Temperature exhibits consistent annual cycling with summer peaks around 25-30°C and winter troughs near 0-5°C. No substantial long-term warming trend is apparent over this seven-year period, though such trends typically require multi-decadal data for statistical significance. The regularity of seasonal oscillations provides a stable baseline for examining pollutant-temperature relationships.

Pollutant Seasonality:

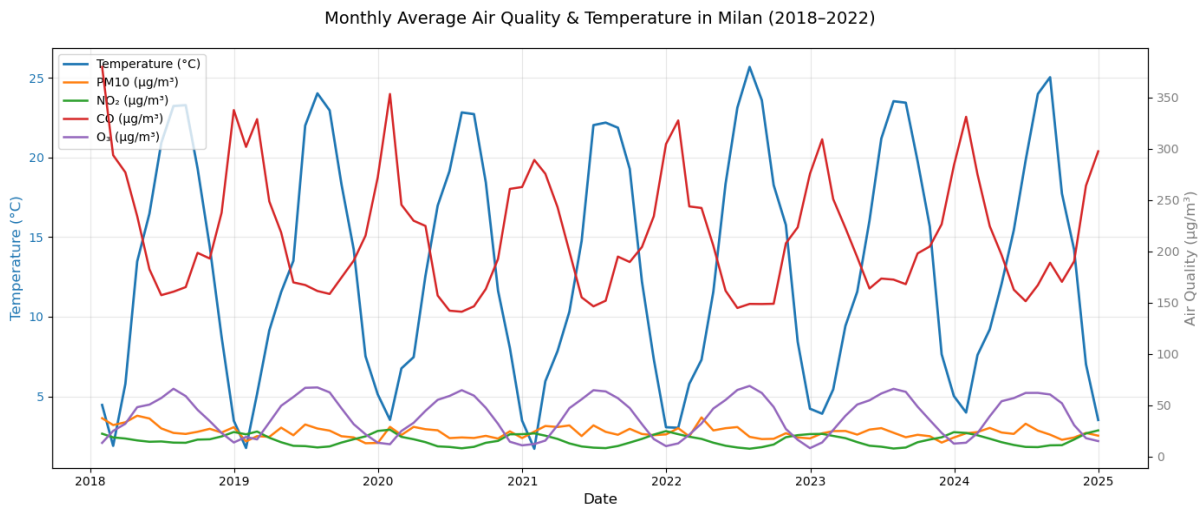
Clear inverse relationships emerge between temperature and primary pollutants (PM_{10}, NO_2, CO). Winter months consistently show elevated concentrations of these species, driven by:

1. **Increased emissions:** Residential heating systems activate, supplementing baseline traffic emissions
2. **Reduced dispersion:** Stable atmospheric conditions and temperature inversions trap pollutants near the surface
3. **Lower mixing heights:** Cold air creates shallow boundary layers, concentrating pollutants in smaller volumes

Carbon monoxide demonstrates the most pronounced winter peaks, frequently exceeding 400 $\mu g/m^3$ and occasionally reaching 600 $\mu g/m^3$ during January-February periods. PM_{10} and NO_2 show similar but less extreme winter elevations.

Conversely, ozone displays opposite seasonality, with summer maxima around 60-80 $\mu g/m^3$ and winter minima below 20 $\mu g/m^3$. This pattern reflects temperature-dependent photochemistry: higher temperatures, increased solar radiation, and longer daylight hours drive ozone formation from NO_x and volatile organic compound (VOC) precursors.

4.3 COVID-19 Impact (2020-2021):



4.3.1: Air Quality and Temperature Patterns (2018–2024)

The pandemic period provides a natural experiment in emission reduction effects. Beginning in March 2020, northern Italy implemented strict lockdown measures that significantly reduced vehicular traffic and industrial activity.

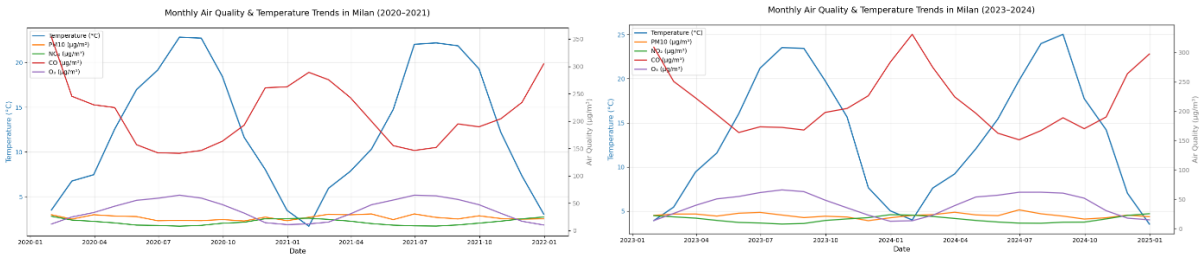
The data reveal measurable air quality improvements during 2020:

- **NO_2 concentrations decreased** noticeably during spring 2020 compared to corresponding periods in 2018-2019, reflecting reduced traffic emissions
- **PM_{10} levels showed modest declines**, though less pronounced than NO_2 due to non-traffic sources (residential heating, dust, regional transport)
- **CO concentrations dropped** during lockdown periods, particularly visible in spring 2020

However, these improvements proved temporary. By late 2021, emission patterns began reverting to pre-pandemic levels, and by 2022-2023, pollutant concentrations had largely returned to pre-COVID patterns. Winter 2023-2024 shows CO peaks comparable to 2018-2019, indicating complete resumption of normal emission activities.

Ozone behavior during lockdowns was more complex. Initial expectations suggested ozone should decrease with reduced NO_x emissions. However, ozone chemistry is non-linear: in NO_x-saturated urban cores, reduced NO_2 can paradoxically increase ozone by reducing the NO-O_3 titration reaction that normally suppresses ozone levels. This phenomenon may explain modest ozone increases observed during some lockdown periods.

4.3.2: The Pandemic Years and After (2020–2025)



The COVID period gave us an unexpected look at what happens when a city suddenly slows down.

Temperature didn't care about lockdowns — it kept cycling normally between seasons, which makes sense since a pandemic doesn't change the weather.

But pollution responded immediately:

When Milan locked down in early 2020, you could see it in the data. NO_2 and PM_{10} dropped noticeably — fewer cars, less industry, cleaner air. It didn't last, though. By late 2021, CO was climbing back up as life returned to normal. By early 2024, we saw CO hitting around 320 µg/m^3 again in January, right back where it started.

Ozone behaved a bit differently during lockdowns, likely because it depends on complex chemical reactions involving NO_2. With less NO_2 around, ozone levels shifted somewhat. But by 2023, ozone was back to its usual summer peaks around 70 µg/m^3.

What we learned:

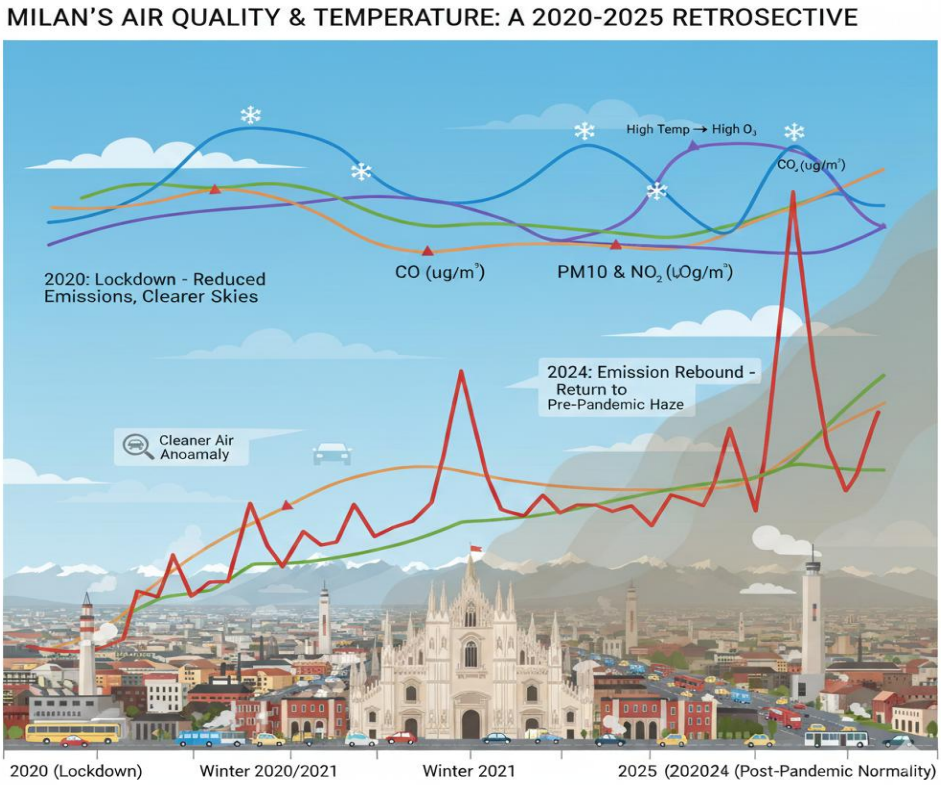
Beyond seasonal patterns, year-to-year variability appears relatively modest, suggesting that meteorological influences on air quality remain stable. No progressive deterioration or improvement trend is evident, indicating that emission patterns and atmospheric conditions have remained relatively constant over the study period, aside from the temporary COVID disruption.

4.3 Summary

Visual exploration of the Milan atmospheric dataset reveals several key characteristics:

1. **Distributional properties** indicate that temperature follows near-normal distribution, while pollutants exhibit right-skewness typical of environmental data
2. **Strong seasonal patterns** dominate the temporal structure, with primary pollutants peaking in winter and ozone maximizing in summer
3. **Temperature emerges as a primary driver** of seasonal air quality variations through its influence on both emissions (heating demand) and atmospheric chemistry (photochemical production)
4. **The COVID-19 period demonstrated** that emission reductions can rapidly improve air quality, but also that improvements require sustained behavioral changes
5. **No long-term trends** are apparent over this seven-year period, though longer time series would be needed to detect climate-driven changes

These visualizations establish the foundation for quantitative correlation and regression analysis in Chapter 5.

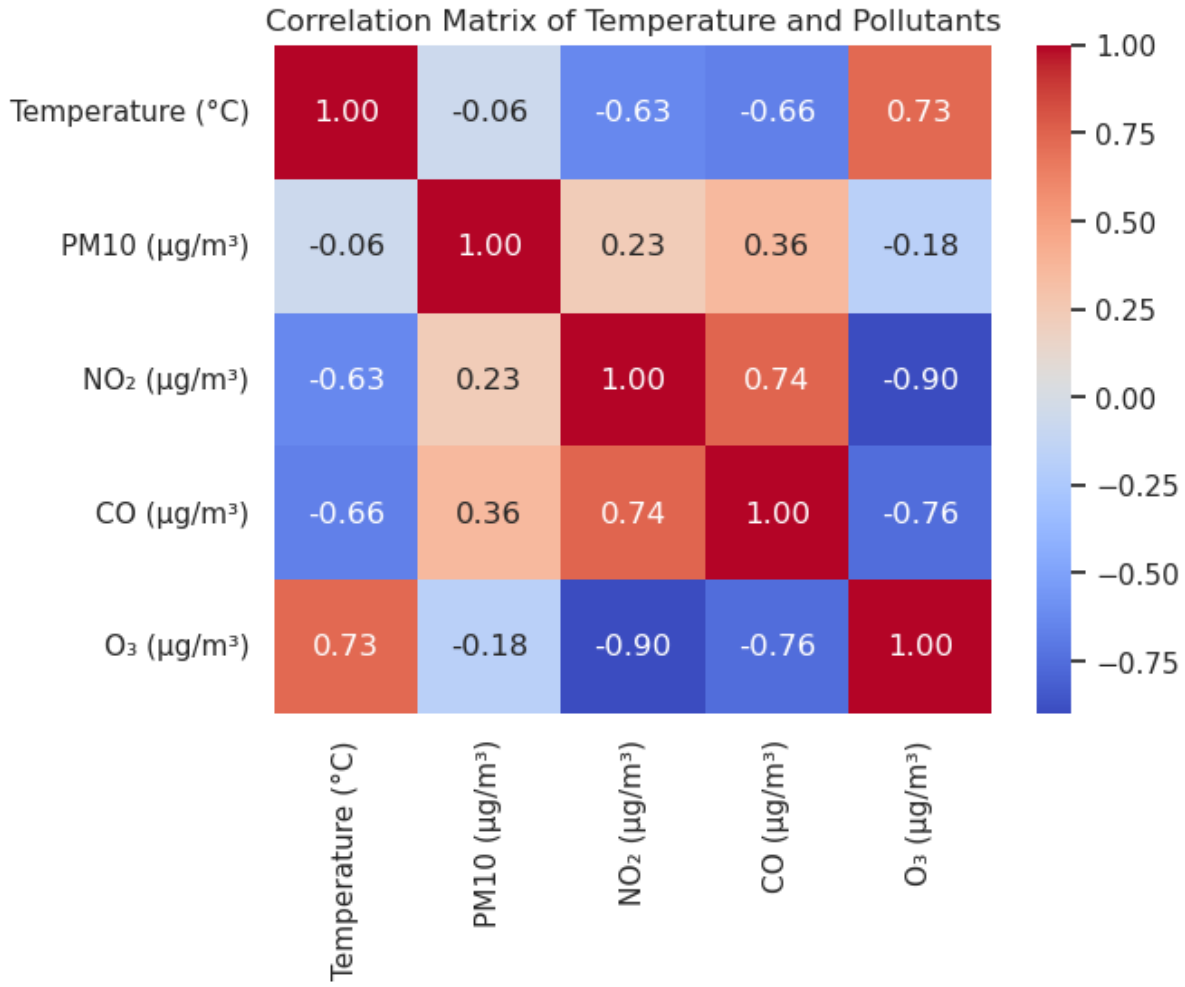


6. TEMPERATURE-POLLUTANT RELATIONSHIP ANALYSIS

Having established the temporal patterns through visualization, this chapter quantifies the relationships between temperature and air pollutants using correlation analysis and linear regression modeling. The goal is to determine the strength and direction of temperature's influence on pollutant concentrations and assess the predictive value of temperature-based models.

5.1 Correlation Matrix Analysis

Pearson correlation coefficients quantify linear relationships between all variable pairs:



Temperature-Pollutant Relationships:

PM_{10} (r = -0.065, p < 0.001): The correlation between temperature and particulate matter is statistically significant but practically negligible. Temperature explains less than 0.5% of PM_{10}

variance ($r^2 = 0.004$). This weak relationship suggests that PM_{10} concentrations are governed primarily by factors other than temperature, including:

- Wind speed and atmospheric turbulence
- Precipitation (wet deposition removal)
- Humidity effects on particle hygroscopic growth
- Specific emission sources (construction, road dust, biomass burning)
- Regional transport from surrounding areas

NO_2 ($r = -0.628$, $p < 0.001$): A strong inverse relationship exists between temperature and nitrogen dioxide. Warmer conditions correlate with significantly lower NO_2 levels, explaining approximately 39% of variance. This relationship reflects multiple mechanisms:

- Reduced heating emissions during warm periods
- Enhanced vertical mixing and dilution at higher temperatures
- Faster chemical oxidation rates converting NO_2 to other species
- Seasonal variation in traffic patterns (though less pronounced than heating)

CO ($r = -0.664$, $p < 0.001$): Carbon monoxide shows the strongest negative correlation with temperature among primary pollutants, accounting for 44% of variance. The relationship is particularly robust because CO is chemically inert on short timescales, making meteorological dilution and emission changes the primary drivers. Cold temperatures increase CO through:

- Intensified residential heating (incomplete combustion)
- Vehicle cold-start emissions (engines less efficient when cold)
- Shallow mixing layers trapping emissions near the surface
- Reduced chemical removal (though CO lifetime is relatively long)

O_3 ($r = +0.729$, $p < 0.001$): Ozone exhibits a strong positive correlation with temperature, explaining 53% of variance—the highest among all pollutants. This relationship is mechanistically driven by temperature-dependent photochemistry. Higher temperatures accelerate ozone formation through:

- Increased reaction rates (photochemical kinetics are temperature-dependent)
- Enhanced biogenic VOC emissions that serve as ozone precursors
- Greater solar radiation intensity correlated with warm conditions
- Reduced NO-O_3 titration when NO_x levels drop in warm seasons

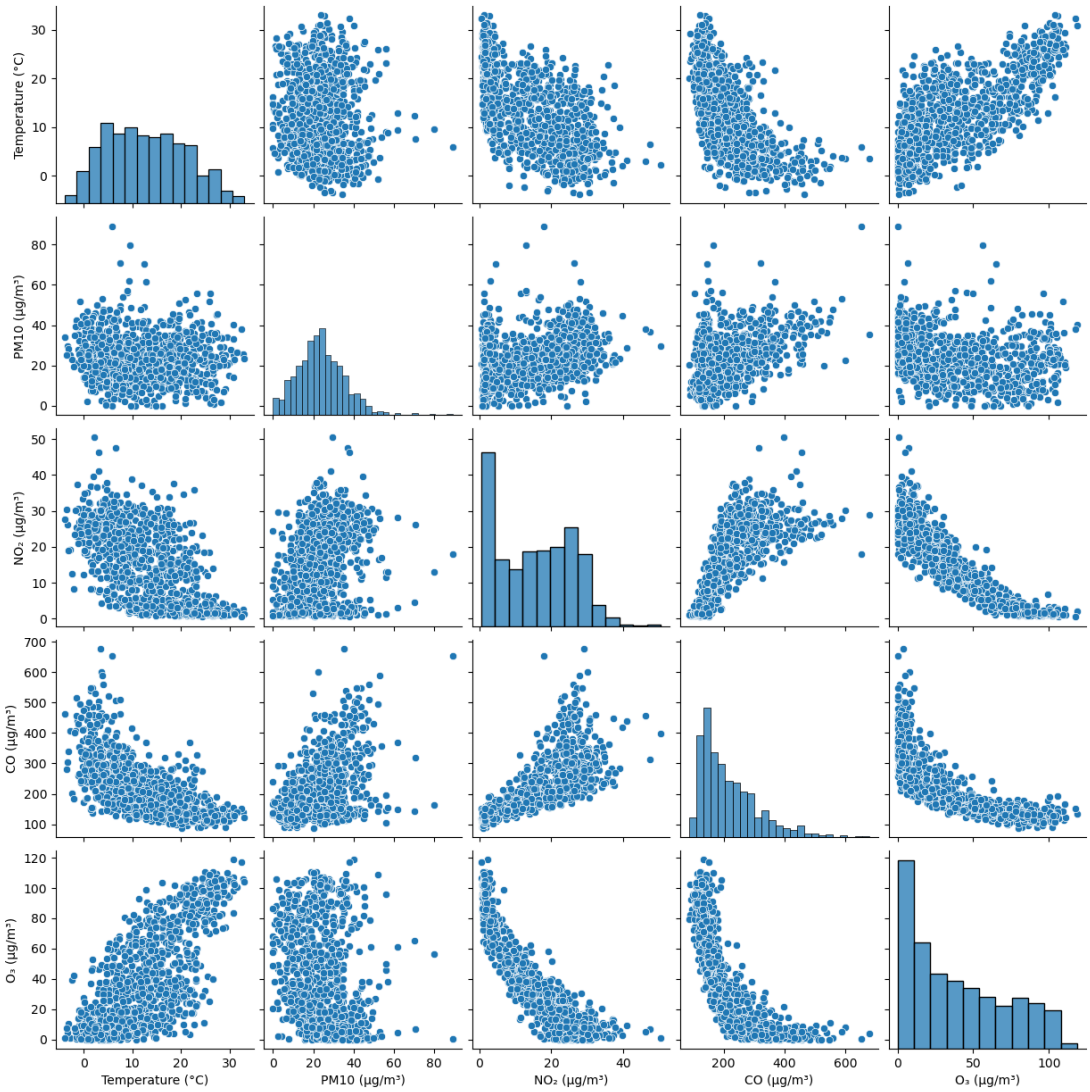
Inter-Pollutant Relationships:

NO_2-CO ($r = +0.876$): The very strong positive correlation indicates shared emission sources, primarily combustion processes. Both species peak simultaneously during winter heating periods and rush-hour traffic, confirming their common origin.

NO_2-O_3 (r = -0.625) and CO-O_3 (r = -0.571): Negative correlations between primary pollutants and ozone reflect their inverse seasonal patterns. Additionally, NO chemically destroys ozone through the $NO + O_3 \rightarrow NO_2 + O_2$ reaction, creating a direct chemical antagonism. This explains why urban centers with high NO_x often have suppressed ozone despite being sources of ozone precursors.

5.2 Pair Plot Analysis

Scatter plots with regression lines illustrate the nature of temperature-pollutant relationships:



Temperature-PM_{10}: The scatter plot reveals a diffuse cloud of points with minimal trend. While a slight negative slope is detectable, the relationship is weak and dominated by scatter, confirming the low correlation coefficient.

Temperature-NO_2: A clear downward trend is evident, with high-density regions showing distinct temperature dependence. Concentrations above 60 µg/m^3 occur almost exclusively below 10°C, while temperatures above 20°C rarely coincide with NO_2 levels exceeding 30 µg/m^3.

Temperature-CO: The negative relationship is visually pronounced and consistent across the temperature range. The regression line shows a steep decline, with CO concentrations halving between winter and summer typical values.

Temperature-O_3: The positive relationship is clearly visible, though not perfectly linear. At temperatures below 5°C, ozone rarely exceeds 30 μg/m^3. Above 25°C, ozone typically ranges from 50-80 μg/m^3. A slight curvature suggests accelerating ozone production at higher temperatures, indicating potential non-linear effects.

5.3 Linear Regression Modeling

To quantify relationships more precisely, simple linear regression models were fitted for each temperature-pollutant pair:

Regression Results Summary:

PM_{10} Model:

- **Slope:** -0.098 μg/m^3 per °C
- **Intercept:** 24.82 μg/m^3
- **R^2:** 0.004
- **p-value:** < 0.001

Despite statistical significance, the model has virtually no predictive power. Temperature explains less than 1% of PM_{10} variability. The negative slope indicates a marginal tendency for lower PM_{10} during warm periods, but this effect is overwhelmed by other factors.

NO_2 Model:

- **Slope:** -0.805 μg/m^3 per °C
- **Intercept:** 26.35 μg/m^3
- **R^2:** 0.394
- **p-value:** < 0.001

Temperature accounts for nearly 40% of NO_2 variance. Each 1°C increase predicts a 0.8 μg/m^3 decrease in NO_2. The model predicts $NO_2 \approx 26$ μg/m^3 at 0°C, declining to approximately 6 μg/m^3 at 25°C (though extrapolation beyond observed ranges should be interpreted cautiously).

CO Model:

- **Slope:** -7.422 μg/m^3 per °C
- **Intercept:** 313.75 μg/m^3
- **R^2:** 0.441
- **p-value:** < 0.001

Temperature explains 44% of CO variance, making it the strongest predictor among primary pollutants. The large slope coefficient reflects CO's dramatic seasonal swing: a 20°C difference between winter and summer predicts a ~148 μg/m^3 change in CO concentration.

O_3 Model:

- **Slope:** +2.914 μg/m^3 per °C
- **Intercept:** 1.50 μg/m^3
- **R^2:** 0.532
- **p-value:** < 0.001

Ozone shows the highest R^2 value (53%), indicating temperature is the dominant meteorological predictor. Each degree of warming predicts a 2.9 μg/m^3 increase in ozone. The model predicts minimal ozone near 0°C, rising to approximately 75 μg/m^3 at 25°C, consistent with observed summer peaks.

5.6 Summary

Quantitative analysis confirms temperature as a significant predictor of air pollutant concentrations in Milan:

PM_{10} shows minimal temperature dependence ($R^2 = 0.004$)

NO_2 and CO exhibit strong inverse relationships ($R^2 \approx 0.40$-0.44)

O_3 demonstrates the strongest positive relationship ($R^2 = 0.53$)

All relationships are statistically significant ($p < 0.001$) and physically interpretable through atmospheric chemistry and meteorology. While simple linear models capture substantial variance for NO_2, CO, and O_3, they do not account for other important factors such as wind patterns, precipitation, weekday/weekend cycles, and long-range transport. Future modeling efforts incorporating these variables could improve predictive accuracy.

The analysis establishes temperature as a primary driver of seasonal air quality patterns in Milan, with important implications for understanding both current pollution dynamics and potential climate change impacts on future air quality.